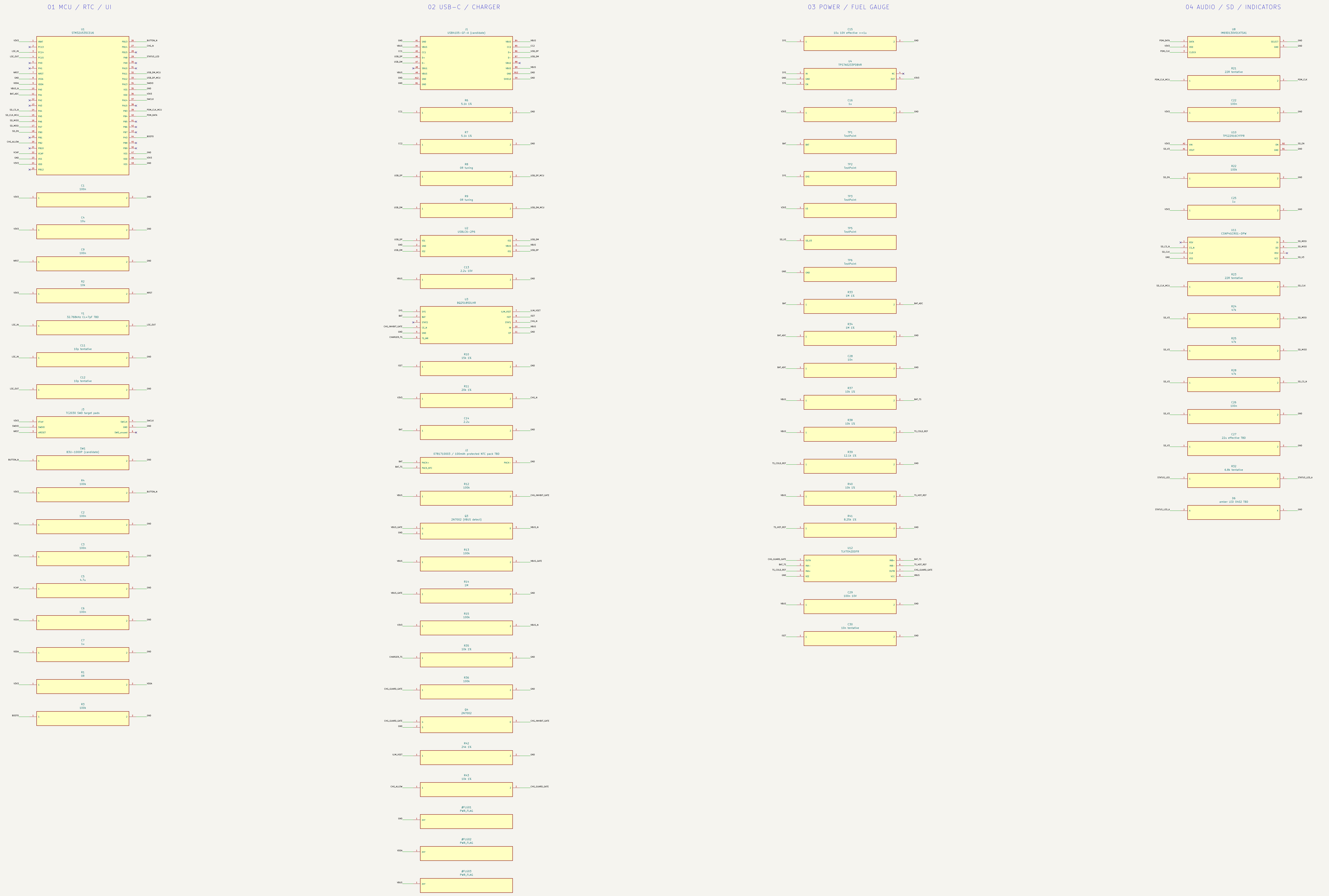
ENGINEERING DRAFT — NOT FOR FABRICATION — See hardware/mini-usb-v9/README.md


01 MCU / RTC / UI
U1
STM32U535CEU6
V3V3 1 VBAT | PB13 26 BUTTON_N
2 PC13 | PB14 27 CHG_N
LSE_IN 3 PC14 | PB15 28
LSE_OUT 4 PC15 | PA8 29 STATUS_LED
5 PH0 | PA9 30
6 PH1 | PA10 31
NRST 7 NRST | PA11 32 USB_DM_MCU
GND 8 VSSA | PA12 33 USB_DP_MCU
VDDA 9 VDDA | PA13 34 SWDIO
VBUS_N 10 PA0 | VSS 35 GND
BAT_ADC 11 PA1 | VDD 36 V3V3
12 PA2 | PA14 37 SWCLK
13 PA3 | PA15 38
SD_CS_N 14 PA4 | PB3 39 PDM_CLK_MCU
SD_CLK_MCU 15 PA5 | PB4 40 PDM_DATA
SD_MISO 16 PA6 | PB5 41
SD_MOSI 17 PA7 | PB6 42
SD_EN 18 PB0 | PB7 43
19 PB1 | PH3 44 BOOT0
CHG_ALLOW 20 PB2 | PB8 45
21 PB10 | PB9 46
VCAP 22 VCAP | VSS 47 GND
GND 23 VSS | VDD 48 V3V3
V3V3 24 VDD | VSS 49 GND
25 PB12
C1 100n: V3V3, GND
C4 10u: V3V3, GND
C9 100n: NRST, GND
R2 10k: V3V3, NRST
Y1 32.768kHz CL=7pF TBD: LSE_IN, LSE_OUT
C11 10p tentative: LSE_IN, GND
C12 10p tentative: LSE_OUT, GND
J3
TC2030 SWD target pads
V3V3 1 VTref | SWCLK 4 SWCLK
SWDIO 2 SWDIO | GND 5 GND
NRST 3 nRESET | SWO_unused 6
SW1 B3U-1000P (candidate): BUTTON_N, GND
R4 100k: V3V3, BUTTON_N
C2 100n: V3V3, GND
C3 100n: V3V3, GND
C5 4.7u: VCAP, GND
C6 100n: VDDA, GND
C7 1u: VDDA, GND
R1 0R: V3V3, VDDA
R3 100k: BOOT0, GND
02 USB-C / CHARGER
J1
USB4105-GF-A (candidate)
GND A1 GND | VBUS B4 VBUS
VBUS A4 VBUS | CC2 B5 CC2
CC1 A5 CC1 | D+ B6 USB_DP
USB_DP A6 D+ | D- B7 USB_DM
USB_DM A7 D- | SBU2 B8
A8 SBU1 | VBUS B9 VBUS
VBUS A9 VBUS | GND B12 GND
GND A12 GND | SHIELD SH GND
GND B1 GND
R6 5.1k 1%: CC1, GND
R7 5.1k 1%: CC2, GND
R8 0R tuning: USB_DP, USB_DP_MCU
R9 0R tuning: USB_DM, USB_DM_MCU
U2
USBLC6-2P6
USB_DP 1 IO1 | IO2 4 USB_DM
GND 2 GND | VBUS 5 VBUS
USB_DM 3 IO2 | IO1 6 USB_DP
C13 2.2u 10V: VBUS, GND
U3
BQ25185DLHR
SYS 1 SYS | ILIM_VSET 7 ILIM_VSET
BAT 2 BAT | ISET 8 ISET
3 STAT2 | STAT1 9 CHG_N
CHG_INHIBIT_GATE 4 CE_N | IN 10 VBUS
GND 5 GND | EP 11 GND
CHARGER_TS 6 TS_MR
R10 15k 1%: ISET, GND
R11 20k 1%: V3V3, CHG_N
C14 2.2u: BAT, GND
J2
0781710003 / 100mAh protected NTC pack TBD
BAT 1 PACK+ | PACK- 3 GND
BAT_TS 2 PACK_NTC
R12 100k: VBUS, CHG_INHIBIT_GATE
Q3
2N7002 (VBUS detect)
VBUS_GATE 1 G | D 3 VBUS_N
GND 2 S
R13 100k: VBUS, VBUS_GATE
R14 1M: VBUS_GATE, GND
R15 100k: V3V3, VBUS_N
R35 10k 1%: CHARGER_TS, GND
R36 100k: CHG_GUARD_GATE, GND
Q4
2N7002
CHG_GUARD_GATE 1 G | D 3 CHG_INHIBIT_GATE
GND 2 S
R42 24k 1%: ILIM_VSET, GND
R43 10k 1%: CHG_ALLOW, CHG_GUARD_GATE
#FLG01 PWR_FLAG: GND 1 pwr
#FLG02 PWR_FLAG: VDDA 1 pwr
#FLG03 PWR_FLAG: VBUS 1 pwr
03 POWER / FUEL GAUGE
C15 10u 10V effective >=1u: SYS, GND
U4
TPS7A0233PDBVR
SYS 1 IN | NC 4
GND 2 GND | OUT 5 V3V3
SYS 3 EN
C16 1u: V3V3, GND
TP1 TestPoint: BAT 1 BAT
TP2 TestPoint: SYS 1 SYS
TP3 TestPoint: V3V3 1 V3
TP5 TestPoint: SD_V3 1 SD_V3
TP6 TestPoint: GND 1 GND
R33 1M 1%: BAT, BAT_ADC
R34 1M 1%: BAT_ADC, GND
C28 10n: BAT_ADC, GND
R37 10k 1%: VBUS, BAT_TS
R38 10k 1%: VBUS, TS_COLD_REF
R39 12.1k 1%: TS_COLD_REF, GND
R40 10k 1%: VBUS, TS_HOT_REF
R41 8.25k 1%: TS_HOT_REF, GND
U12
TLV7042DDFR
CHG_GUARD_GATE 1 OUTA | INB+ 5 BAT_TS
BAT_TS 2 INA- | INB- 6 TS_HOT_REF
TS_COLD_REF 3 INA+ | OUTB 7 CHG_GUARD_GATE
GND 4 VEE | VCC 8 VBUS
C29 100n 10V: VBUS, GND
C30 10n tentative: ISET, GND
04 AUDIO / SD / INDICATORS
U8
IM69D130V01XTSA1
PDM_DATA 1 DATA | SELECT 4 GND
V3V3 2 VDD | GND 5 GND
PDM_CLK 3 CLOCK
R21 22R tentative: PDM_CLK_MCU, PDM_CLK
C22 100n: V3V3, GND
U10
TPS22916CYFPR
V3V3 A2 VIN | ON B2 SD_EN
SD_V3 A1 VOUT | GND B1 GND
R22 100k: SD_EN, GND
C25 1u: V3V3, GND
U11
CSNP4GCR01-DPW
1 RSV | DI 5 SD_MOSI
SD_CS_N 2 CS_N | DO 6 SD_MISO
SD_CLK 3 CLK | RSV 7
GND 4 VSS | VCC 8 SD_V3
R23 22R tentative: SD_CLK_MCU, SD_CLK
R24 47k: SD_V3, SD_MOSI
R25 47k: SD_V3, SD_MISO
R28 47k: SD_V3, SD_CS_N
C26 100n: SD_V3, GND
C27 22u effective TBD: SD_V3, GND
R32 6.8k tentative: STATUS_LED, STATUS_LED_A
D6
amber LED 0402 TBD
STATUS_LED_A 2 A | K 1 GND

Unrouted PCB / no manufacturing release

Sheet: /
File: pendant-mini-usb-v9.kicad_sch

Title: Pendant Recorder — Omi-derived electrical design

Size: A0 | Date: 2026-10-25 | **Rev: mini-usb-v9 DRAFT**
KiCad E.D.A. 10.0.7 | Id: 1/1